

SIQI WANG

Hefei, China | +86 15285753543

peking_wsq@mail.ustc.edu.cn | [Personal Website](#)

EDUCATION

B.S. in Physical Chemistry, University of Science and Technology of China Sep 2023 – Present

School of Chemistry and Materials Science | Minor: Precision Intelligent Chemistry (AI for Chemistry)

GPA: 3.85/4.30 | Overall Rank: **7/135** | Major Rank: **2/23**

Selected Coursework: Statistical Mechanics (88), Physical Chemistry III (95), Quantum Physics (93), Complex Analysis (90), Computer Programming (96), Mathematical Analysis I (90) & II (92), Linear Algebra (93)

RESEARCH EXPERIENCE

Nonequilibrium Modeling of ATP-Driven Biomolecular Systems Jul 2026 – Present

École Polytechnique Fédérale de Lausanne | Advisor: Prof. Paolo De Los Rios *Lausanne, Switzerland*

- Implemented and analyzed a kinetic model of the ATP-driven Hsp70 chaperone cycle to investigate how chemical energy consumption reshapes substrate binding and steady-state populations beyond equilibrium constraints.
- Numerically solved the four-state reaction network under mass-conservation and thermodynamic-consistency constraints to quantify steady-state populations, effective binding affinity, and probability currents across ATP/ADP and hydrolysis-rate regimes.
- Analyzed ultra- and infra-affinity regimes through parameter scans and cycle-flux calculations, connecting nonequilibrium thermodynamic driving to changes in binding behavior and underlying kinetic mechanisms.
- Explored phase-field modeling of biomolecular condensates and potential extensions coupling nonequilibrium biochemical reactions to phase-separation dynamics.

Computational Design of Near-Infrared Fluorescent Proteins Jul 2025 – Mar 2026

University of Science and Technology of China | Advisor: Prof. Jie Zhang *Hefei, China*

- Led a computational protein-design project aimed at designing fluorescent proteins incorporating exogenous small-molecule chromophores.
- Developed an HPC-based design pipeline integrating binding-site motif construction, RFdiffusion3 for constrained backbone generation, LigandMPNN for sequence design, and AlphaFold for independent structure prediction.
- Evaluated structural consistency between designed and predicted structures using RMSD-based metrics and structure-based filtering criteria, yielding a preliminary set of candidate designs for subsequent molecular dynamics and experimental validation.

Molecular Dynamics Under Photoexcitation Jul 2025 – Aug 2025

Westlake University | Advisor: Prof. Gu Bing *Hangzhou, China*

- Received training in quantum chemistry and theoretical modeling of molecular photophysical processes, translating theoretical concepts into computational workflows.
- Performed electronic-structure calculations using PySCF and Python; constructed molecular models and analyzed computed energies and potential-energy profiles for molecular systems.

LEADERSHIP AND ACTIVITIES

Teaching Assistant, Analytical Chemistry I Sep 2025 – Present

- Graded weekly homework using a standardized rubric; provided written feedback on common errors.
- Managed course announcements and uploaded problem sets to the course platform.
- Led weekly tutorial or recitation sessions for a sophomore-level Analytical Chemistry I course online; mentored students on problem-solving strategies and exam preparation.

- Awarded Third Prize in Anhui Provincial Division.

AWARDS AND HONORS

Outstanding Project (University-Level), National Undergraduate Innovation Training Program, USTC	Mar 2026
Jiaxi Lu Elite Scholarship (Category A) (Top 2%), USTC	Oct 2025
Jiaxi Lu Scholarship (Top 5%), USTC	Oct 2025
Outstanding Student Silver Scholarship (Top 10%), USTC	Nov 2024
Jiaxi Lu Elite Scholarship (Category A) (Top 2%), USTC	Oct 2024
Outstanding Freshman Scholarship, USTC	Oct 2023

SKILLS

Programming: Python (NumPy, SciPy, Matplotlib), C
Scientific Computing & Modeling: PySCF, Gaussian, GROMACS, PyRosetta
Tools: PyMOL, Origin, LaTeX
Languages: Chinese (Native), English (TOEFL iBT: 6.0/6.0; comparable score: 118/120)